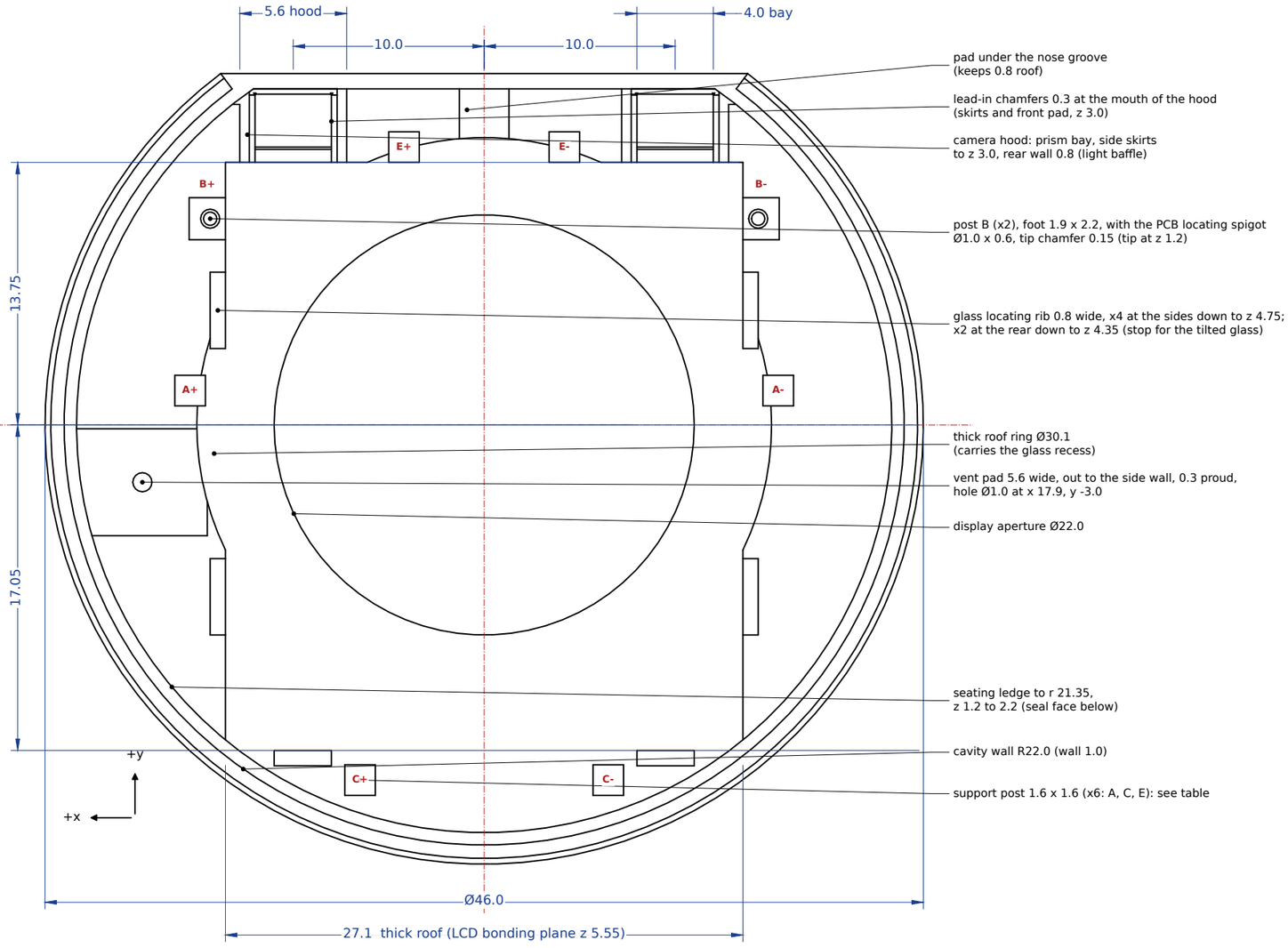
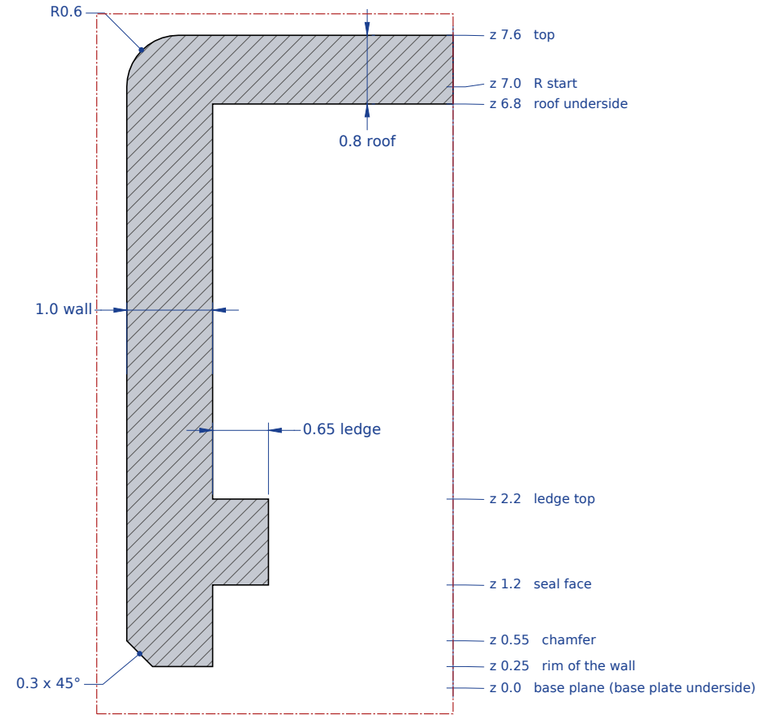
VIEW FROM BELOW - into the cavity 3.5:1 (+x is to the LEFT, visor at the top)



DETAIL E - wall, ledge and roof on the arc (section y = 0 at x = -23.0) 14:1



Post	x from	x to	y from	y to	foot	spigot centre
A-	-16.2	-14.6	1.0	2.6	1.6 x 1.6	-
A+	14.6	16.2	1.0	2.6	1.6 x 1.6	-
B-	-15.45	-13.55	9.7	11.9	1.9 x 2.2	(-14.35, 10.8)
B+	13.55	15.45	9.7	11.9	1.9 x 2.2	(14.35, 10.8)
C-	-7.3	-5.7	-19.4	-17.8	1.6 x 1.6	-
C+	5.7	7.3	-19.4	-17.8	1.6 x 1.6	-
E-	-5.0	-3.4	13.75	15.35	1.6 x 1.6	-
E+	3.4	5.0	13.75	15.35	1.6 x 1.6	-

All posts: PCB top (z 1.8) to the roof. Feet lie inside the YAML keep-out squares (check 12); posts B: 2.2 square, foot cut back to |x| >= 13.55 for the LCD glass (D29).

NOTES

1. Material HP 3D High Reusability PA 12, Multi Jet Fusion, dyed black, chemical vapor smoothing. Print roof-down (cosmetic top on the bed side, cavity open upward).
2. Minimum wall 0.8 mm by design. Recorded exceptions: the 0.525 mm strip behind each window recess (D04, accepted), the feet of chamfered edges (side wall 0.7 at the rim, hood rear wall 0.68, hood skirts 0.5 at their lead-in chamfer). See export/check_report.md, check 5.
3. No closed void and no cavity without a 2 mm escape path (check 6). Depowder the two camera bays, the visor openings and the Ø1.0 vent hole with care.
4. Critical-to-function: bonding plane z 5.55 (LCD), recess floor z 6.35 (cover glass), seal face z 1.2, post feet z 1.8 (all in one plane +-0.1), window recesses, ledge radius 21.35 (go gauge), bay roof 5.0 above the post feet (go gauge). REQUIRED +-0.1, measured from the post feet: seal face 0.6 below them (it sets the seal joint), rim 1.55 below them (it sets the rim relief). The bureau's general tolerance is wider: first-article inspection of these is a release gate.
5. Drop-in fits are 0.25 mm nominal. The two spigots under posts B (28.7 apart; Ø1.0 in a Ø1.15 PCB hole at x = 14.35 and in a 1.15 x 1.8 slot at x = -14.35) locate the PCB: go gauge Ø1.10 at first article (D19).
6. Camera hoods stay inside the YAML keep-out zones (camera x +-2.85, y 13.7 to 17.3); skirts end at z 3.0 (1.2 above the PCB).
7. The side wall ends 0.25 mm above the base plane (rim relief): z values on this drawing are heights above the base plane, the tub itself is 7.35 tall. No key lug: the base plate has a plain edge and the seal face runs unbroken round the D. Vent: pad 0.3 proud, drain grooves 0.6 x 0.3 on the top face.
8. Rev D: spigots moved from posts E to posts B; posts E are plain. Window recesses unchanged (5.5 x 0.55 deep, z 4.2 to the top) for 5.0 x 3.25 x 0.4 glass windows seated on the recess floor. The two rear glass ribs reach down to z 4.35 (the free height above the PCB): the tilted LCD glass is pushed back against them. Nothing has been built.

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

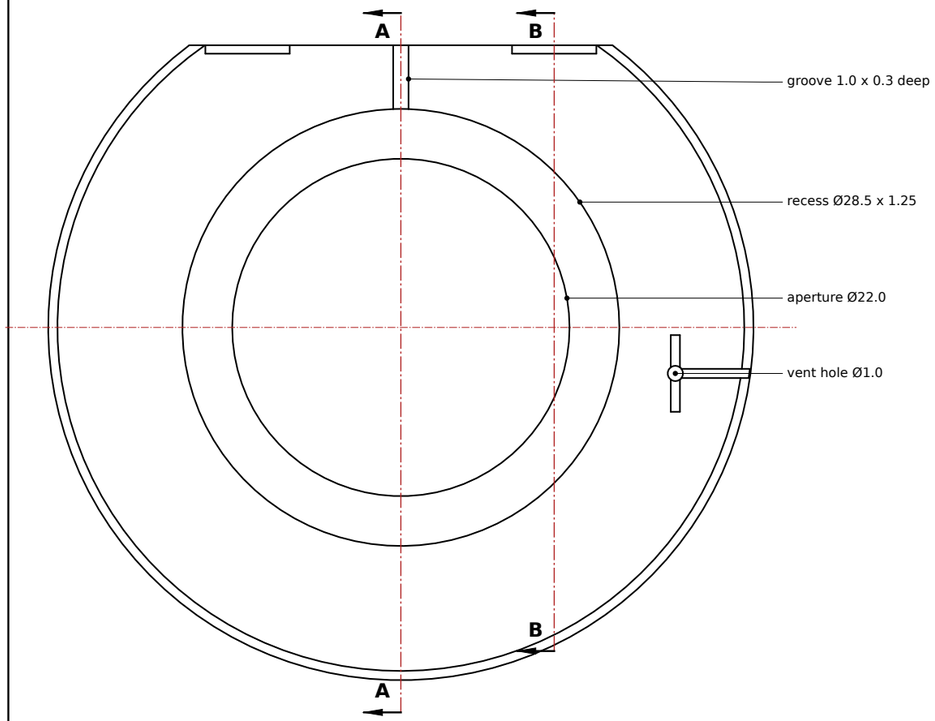
TITLE
GL1-010 Tub (top shell)

PROJECT / PART
GreenLine GL-1 electronic ball marker - tub

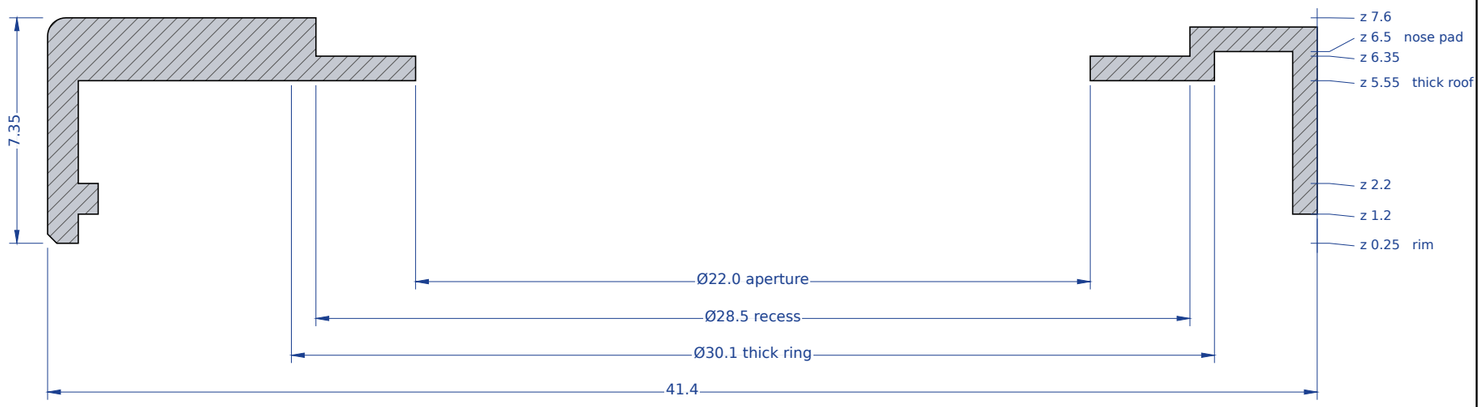
MATERIAL / FINISH	PROCESS
PA12 (HP 3D HR PA 12), dyed black	HP Multi Jet Fusion + vapor smoothing
GENERAL TOLERANCE	SCALE (A3) / UNITS
printed: +-0.3 or +-0.3 % (bureau MJF), as printed	3.5:1 / 14:1 mm

STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	D-EVT1 / D-EVT1	1 of 2 2026-11-11

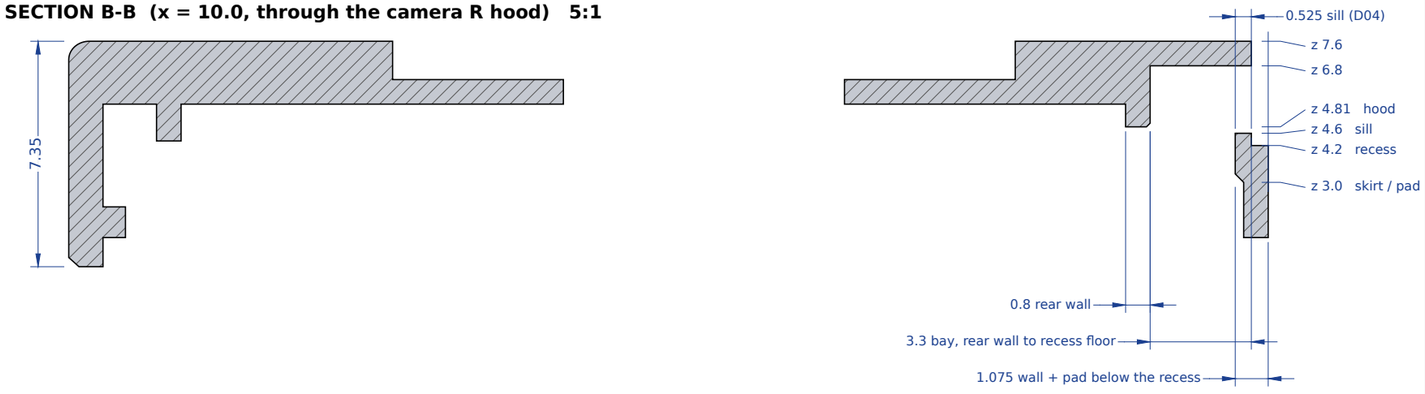
TOP VIEW 2.5:1 (section planes A-A and B-B)



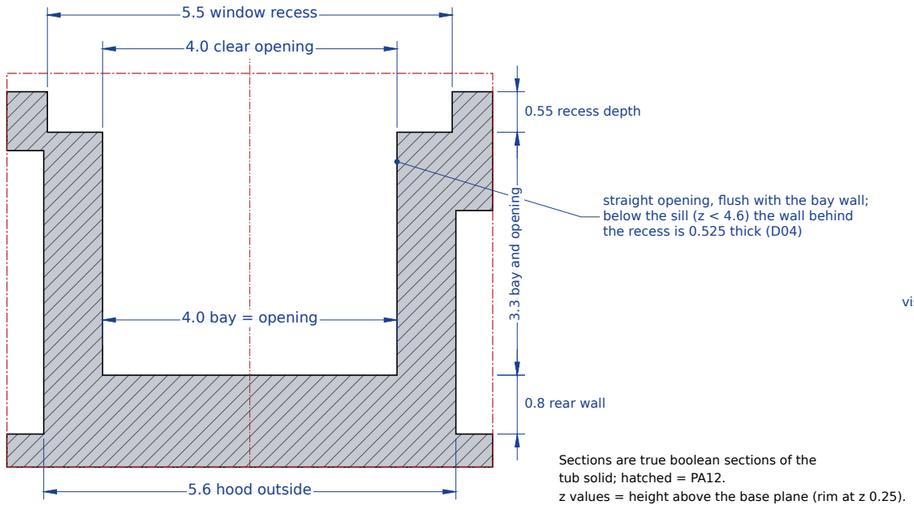
SECTION A-A (x = 0) 5:1



SECTION B-B (x = 10.0, through the camera R hood) 5:1

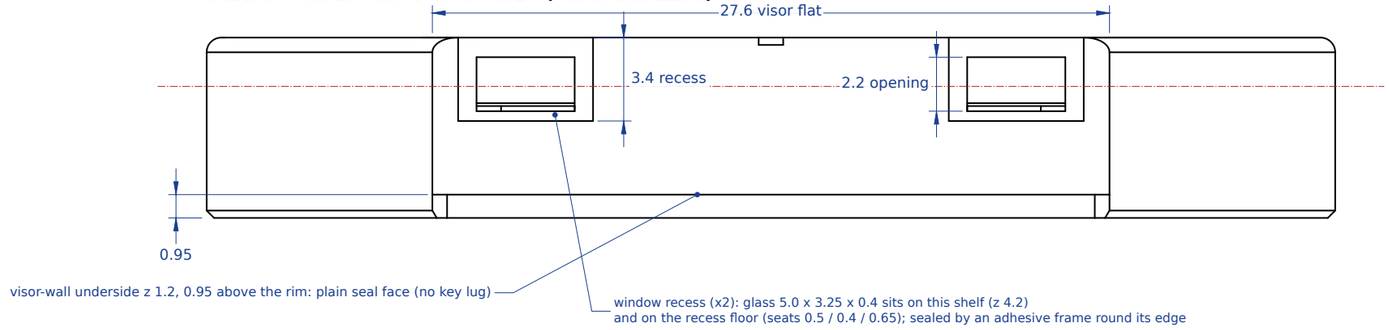


DETAIL G - camera R opening, section at z = 5.61 (seen from above) 12:1



Sections are true boolean sections of the tub solid; hatched = PA12.
z values = height above the base plane (rim at z 0.25).

VIEW F - on the visor face 4:1 (+x to the LEFT)



TITLE: **GL1-010 Tub (top shell)**		
PROJECT / PART: GreenLine GL-1 electronic ball marker - tub		
MATERIAL / FINISH: PA12 (HP 3D HR PA 12), dyed black	PROCESS: HP Multi Jet Fusion + vapor smoothing	
GENERAL TOLERANCE: printed: +-0.3 or +-0.3 % (bureau MJF), as printed	SCALE (A3) / UNITS: 2.5:1 / 5:1 / 12:1 mm	
STATUS: **PROVISIONAL - not released**	DWG REV / PARAMS: D-EVT1 / D-EVT1	SHEET / DATE: 2 of 2 2026-11-11

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.